INTERNET OF THINGS (IoT)

Midterm Question Paper – Easy, Simple Answers | Course Code: CSE0612405

Exam tip: These answers are written in simple language. For 2-mark questions, write the definition and key point. For 4-mark questions, explain the points and draw the diagram where asked.

SET A — Total Marks: 10 (2 + 4 + 4)

Q1. What is IoT? Write the main goal of IoT. [2]

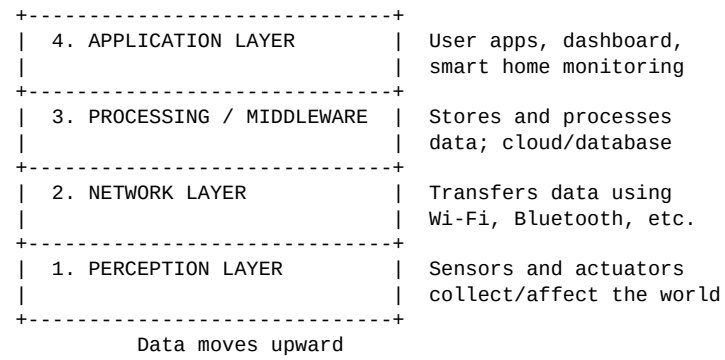
Internet of Things (IoT) is a system in which physical objects (such as sensors, machines, vehicles and home appliances) are connected to the internet. They collect and exchange data with other devices.
Main goal: To make devices smart so they can monitor, communicate and perform tasks automatically with little human effort.

Q2. What is an AVR microcontroller? Write its history, why it is special, and name its types. [4]

AVR microcontroller: AVR is an 8-bit RISC microcontroller family originally developed by Atmel. It is used to control electronic devices and embedded systems.
History: It was developed by two Norwegian engineers, Alf-Egil Bogen and Vegard Wollan, and introduced by Atmel in the 1990s. Atmel later became part of Microchip Technology.
Why is it special? It has a simple instruction set, fast operation, low power use, built-in memory and input/output pins. It is also easy to program and widely used in education and embedded projects.
Common AVR types/families: tinyAVR, megaAVR, XMEGA and newer AVR families such as AVR Dx. **ATmega32** is an example of megaAVR.

Q3. Draw and explain the four-layer IoT architecture. [4]

The four-layer architecture explains how data moves from a real-world object to an application.



1. **Perception layer:** Sensors collect information such as temperature, light or movement. Actuators perform actions.
2. **Network layer:** Sends data between devices, gateways and servers using Wi-Fi, Bluetooth, cellular networks, etc.
3. **Processing/Middleware layer:** Stores, filters and processes the data. It may use cloud platforms and databases.
4. **Application layer:** Provides services to users, such as smart homes, smart health and smart agriculture.

SET B — Total Marks: 10 (2 + 4 + 4)

Q1. Differentiate between physical things and virtual things with one example each. [2]

Physical things: Real objects that can be touched. They have a physical form and may contain sensors or actuators.
Example: A temperature sensor or smart fan.
Virtual things: Digital objects or software representations that exist in a computer system. They cannot be physically touched. **Example:** A digital model of a smart fan shown in an app.

Q2. Explain I/O programming of AVR (ATmega32) using DDRx, PORTx and PINx registers with examples. [4]

AVR uses three main registers to control an I/O port:
• **DDRx (Data Direction Register):** Selects the pin direction. Bit 1 = output; bit 0 = input.

• **PORTx (Port Register):** For an output pin, writes HIGH (1) or LOW (0). For an input pin, writing 1 enables the internal pull-up resistor.
• **PINx (Pin Input Register):** Reads the actual logic level present at the pins.
Example: Set PB0 as output and turn on an LED:

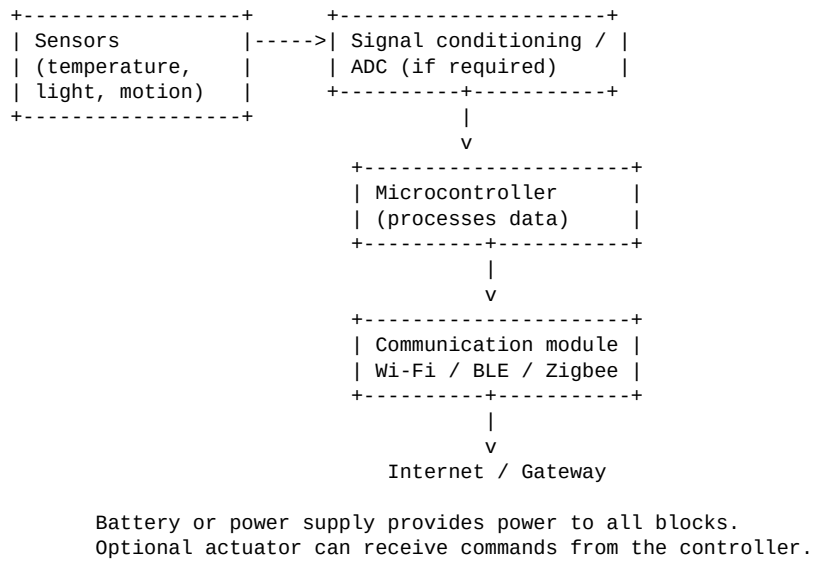
```
DDRB |= (1 << PB0);       // PB0 is output
PORTB |= (1 << PB0);      // PB0 HIGH: LED ON
PORTB &= ~(1 << PB0);     // PB0 LOW: LED OFF

DDRB &= ~(1 << PB1);      // PB1 is input
if (PINB & (1 << PB1)) {  // Read PB1
    // Pin is HIGH
}
```

Here, DDRB controls the direction, PORTB writes an output value, and PINB reads the input value. The letter x means the port name, such as A, B, C or D.

Q3. Draw the block diagram of an IoT sensor node and explain its parts. [4]

An IoT sensor node collects information from the environment, processes it and sends it to another device or the internet.



1. **Sensor:** Measures physical conditions such as temperature, humidity or movement.
2. **Signal conditioning/ADC:** Makes the sensor signal suitable for the controller. ADC converts an analog signal into digital data when needed.
3. **Microcontroller:** Reads and processes sensor data and controls the node.
4. **Communication module:** Sends data to a gateway or cloud using Wi-Fi, Bluetooth or another network.
Power supply: Battery or adapter provides energy to the node.

SET C — Total Marks: 10 (2 + 4 + 4)

Q1. Write two differences between a microprocessor and a microcontroller with examples. [2]

Microprocessor: Mainly contains the CPU. It usually needs external RAM, storage and input/output devices.
Example: Intel Core processor.
Microcontroller: Contains a CPU, memory and input/output peripherals on one chip. It is designed to control a specific device. **Example:** ATmega32.
Two differences: (1) A microprocessor generally needs more external components; a microcontroller has many components built in. (2) Microprocessors are common in computers; microcontrollers are common in embedded systems.

Q2. Explain the types of IoT system (Level 1 to Level 6) with a simple diagram. [4]

The six IoT levels describe systems from a simple single device to a large system with many devices and cloud services. A common textbook classification is:

```
LEVEL 1: One device -> local processing / local use
LEVEL 2: One device -> cloud storage -> application
LEVEL 3: One device -> cloud -> data analytics
LEVEL 4: Many devices -> cloud storage and analytics
LEVEL 5: Many end nodes -> coordinator/gateway -> cloud
LEVEL 6: Large, independent devices/systems -> cloud
         analytics, monitoring and control
```

Level 1: One device performs sensing, processing and action locally. Suitable for a simple, low-cost system.
Level 2: One device sends data to the cloud for storage; users access it through an application.
Level 3: One device sends data to the cloud, where more powerful analytics are performed.
Level 4: Multiple devices collect data and send it to the cloud for storage and analysis.
Level 5: Many end devices communicate through a coordinator or gateway, which connects them to the cloud.
Level 6: A large and complex IoT system has many independent devices or subsystems. Cloud services perform large-scale analysis and may support central monitoring and control.

Q3. Draw and explain the seven-layer IoT World Forum (IoTWF) architecture. [4]

The IoT World Forum (IoTWF) model has seven layers. Data generally moves upward, while commands can move downward.

```
+---------------------------------+
| 7. COLLABORATION & PROCESSES      | People use data; take action
+---------------------------------+
| 6. APPLICATION                    | IoT apps and dashboards
+---------------------------------+
| 5. DATA ABSTRACTION               | Organizes and makes data usable
+---------------------------------+
| 4. DATA ACCUMULATION              | Stores data (data at rest)
+---------------------------------+
| 3. EDGE COMPUTING                 | Filters and analyzes near device
+---------------------------------+
| 2. CONNECTIVITY                   | Moves data across networks
+---------------------------------+
| 1. PHYSICAL DEVICES & CONTROLLERS | Sensors, actuators, machines
+---------------------------------+
 Security and management apply across all layers.
```

1. Physical devices and controllers: Sensors, machines and actuators collect data or perform actions.
2. Connectivity: Transfers data through wired or wireless networks.
3. Edge computing: Processes data near the device to reduce delay and network traffic.
4. Data accumulation: Stores incoming data for later use.
5. Data abstraction: Organizes and combines data so applications can use it easily.
6. Application: Provides monitoring, control and analysis software.
7. Collaboration and processes: People and business processes use the information to make decisions and take action.
Note: Security and management are important across all seven layers.

SET D — Total Marks: 10 (2 + 4 + 4)

Q1. What is sensor calibration? Why do we need it? [2]

Sensor calibration is the process of comparing a sensor's readings with a known standard and adjusting it so that it gives correct measurements.
Why is it needed? It improves accuracy, reduces measurement errors and helps keep readings reliable over time.

Q2. Describe any four main challenges of IoT implementation. [4]

1. **Security:** Connected devices may be attacked or hacked if they are not protected.
2. **Privacy:** IoT devices can collect personal data, so that data must be kept private.
3. **Connectivity:** Devices need a stable network to send and receive information. Weak internet can interrupt the system.
4. **Interoperability:** Devices from different companies may use different standards and may not work together easily.

Other challenges: Power consumption, handling large amounts of data, high installation cost and system maintenance.

Q3. Differentiate parallel and serial communication. Also name and explain the common interfacing hardware (USB, GPIO, I2C, SPI, UART, RS232). [4]

Parallel communication: Sends several bits at the same time using multiple data wires. It can transfer data quickly over short distances, but needs more wires.
Serial communication: Sends bits one after another through one or a few wires. It needs fewer wires and is commonly used for communication between devices.
Common interfaces:

Interface	Simple explanation
USB	Universal Serial Bus. Connects devices such as keyboards, phones and storage devices; can also provide power.
GPIO	General-Purpose Input/Output pins. A controller can read a digital input or set a digital output, such as switching an LED.
I2C	A two-wire serial bus using SDA (data) and SCL (clock). It can connect multiple addressed devices.
SPI	A fast synchronous serial interface commonly using SCLK, MOSI, MISO and chip-select (CS) lines.
UART	A serial communication method that commonly uses TX and RX lines. It is asynchronous, so no shared clock line is required.
RS232	A serial communication standard that defines signal voltage levels and connections, often used between computers and older devices.

Remember: I2C uses two main signal wires; SPI commonly uses four signal types; UART uses TX and RX; GPIO is for general digital input/output.